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Preparation of hydroxycinnamic acid and hydroxycinnamate from herbal lignin in aprotic solvent under mild conditions

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ABSTRACT

Herbaceous (grass-type) lignin is characterized by a high content of hydroxycinnamic acid, which are typically ester-linked to the γ-position of the lignin side chains. These hydroxycinnamate units serve as valuable precursors for a variety of fine chemicals. This study constructed a mild and efficient MeOK-DMC/THF system to cleave ester bonds in herbaceous lignin. The effects of solvent composition, alkali metal alkoxide type, reaction temperature, and biomass feedstock were systematically investigated. Under optimized conditions, bamboo lignin yielded 15.6 wt% hydroxycinnamic acid and hydroxycinnamate with a selectivity of 84.5 %, while bagasse lignin achieved a higher yield of 20.8 wt% and selectivity of 97.7 %. Model compound studies confirmed that hydroxycinnamic acid and hydroxycinnamate are generated through the γ-O ester bond cleavage reaction of hydroxycinnamic acid units. This work provides a green and scalable strategy for the valorization of renewable lignin into high-value aromatic compounds, contributing to sustainable biorefinery development.

1. Introduction

With growing environmental concerns and increasing demand for petrochemicals, the development of sustainable methods to produce fuels and chemicals from sustainable resources has become an urgent global priority (Sun et al., 2018, Rath et al., 2024). Biomass utilization plays a vital role in addressing the mitigating environmental issues associated with their excessive use (Ye et al., 2021, Chen et al., 2021). Lignocellulosic biomass, mainly composed of cellulose, hemicellulose, and lignin, is widely regarded as a promising alternative to fossil resources for the production of high-value fuels and chemicals. Among these components, lignin is particularly valuable due to its rich aromatic structure (Rinaldi et al., 2016, Liao et al., 2020).

As the only renewable aromatic polymer in nature, lignin comprises three primary monolignol precursors: *p*-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol, which form the *p*-hydroxyphenyl (H) guaiacyl (G), and sinapyl (S) units (Liu et al., 2020, Rinaldi et al., 2016, Ralph et al., 2019). Recent studies have revealed some new features about the grass lignin. For example, tricin, a valuable flavonoid, has been identified as the important monomer for lignification in grass. Besides, substantial amount of *p*-coumarate and ferulate ester have been found to incorporate into grass lignin. (Ralph, 2009, Kumar et al., 2020). Specifically, *p*-coumaric acid (*p*CA) and ferulic acid (FA) are esterified at the C_γ of β-O-4 linkages. These hydroxycinnamates are not only valuable aromatic compounds but also represent a unique route for the valorization of herbaceous lignin.

Hydroxycinnamic acid and its derivatives (HADs) serve as versatile intermediate in the food, spices, cosmetics, and pharmaceutical industries (Wu et al., 2025, Kuang et al., 2025, Yang et al., 2023, Kabat et al., 2024). These compounds exhibit diverse bioactivities, including anti-tumor(Ceylan et al., 2025), anti-inflammatory (Tan et al., 2024), antioxidant(Perinelli et al., 2022, Malarz et al., 2023) and antifungal (Ojeda-Hernandez et al., 2023). However, conventional chemical synthesis of hydroxycinnamate often relies on fossil-based feedstocks, noble metal catalysts, or corrosive acid catalysts, which are associated with high costs, complex procedures, and environmental concerns (Rana and Jonnalagadda, 2017, Wu et al., 2022).

The abundant hydroxycinnamic acid and hydroxycinnamate on grass

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lignin offers an alternative source for the production of natural and renewable Hydroxycinnamic acid and its derivatives. Sequential alkaline and acid hydrolysis has been applied to release 0.79 % of FA and 4.38 % of *p*CA from sugarcane bagasse (Li et al., 2019). Sodium carbonate treatment was also effective to for hydroxycinnamic acid extraction, yielding 15.6 ± 1.9 mg CAE g^{-1} DM polyphenol products extracted from cotton stalks, which are particularly enriched in two hydroxycinnamate derivatives, FA and *p*CA. (Papadimitriou et al., 2024). Despite these advances, alkaline conditions can promote side reactions such as aldol condensation, leading to decreased yields and selectivity (Liu et al., 2021).

Recent progresses offer new opportunities for more sustainable and selective production of hydroxycinnamates from grass lignin (Martinkova et al., 2023). For instance, Timokhin et al. achieved 4.8 wt % crude yield of *p*CA using γ-valerolactone as the solvent from corn stover (Timokhin et al., 2020). Shi et al. and Wang et al. employed $ZnMoO_4$/MCM-41 catalyst to selectively release coumaric acid and ferulic acid from lignin in corncob lignin (Shi et al., 2023, Wang et al., 2018). However, many of these systems still suffer from harsh reaction conditions or catalyst instability. Metal based ionic liquids, such as [Bmim][$FeCl_4$], have been shown to selectively cleave *p*CA and transformed it into methyl *p*-coumarate (M*p*C) with a yield of 71.1 mg g^{-1}(Li et al., 2018). Nevertheless, the high cost of ionic liquids and difficulties of product separation remain major obstacles to their practical application.

In previous study, our group have developed an EtONa-THF/DMSO system capable of cleaving lignin β-O-4 ether bonds at low temperatures, yielding benzoic acid derivatives (Zeng et al., 2023). This reaction system also exhibited capability for ester cleavage. Building upon this, we have now designed a mixed DMC-THF solvent system incorporating alkali metal alkoxides to selectively cleave ester bonds under mild conditions. This approach enables the efficient production of hydroxycinnamic acid and hydroxycinnamate from herbaceous lignin. We systematically investigated the influence of solvent composition, base type, temperature, and lignin source on hydroxycinnamic acid and hydroxycinnamate yield and selectivity. Compared with traditional NaOH-based systems, our method operates under milder conditions while achieving higher yields and improved selectivity.

2. Materials and methods

2.1. Materials

Moso bamboo was sourced from Yong'an He'an Industry and Trade Co., Ltd., Sanming, China, and bagasse was obtained from Yunnan, China. Acetone (>98 %), acetic acid (CH_3COOH, ≥99.5 %), hydrochloric acid (HCl, 36 %) were purchased from Guangzhou Chemical Reagent Factory, Guangzhou, China. Sodium hydroxide (NaOH, ≥99.7 %) were purchased from Fuchen Chemical Reagent Co., Ltd., Tianjin , China. Dimethyl carbonate (DMC,99 %), methanol lithium (MeOLi, 98 %), sodium methoxide (MeONa, 98 %), pyridine (≥99.9 %), potassium methoxide (MeOK, 95 %), bis (trimethylsilyl) trifluoroacetamide (BSTFA, 98 %), naphthalene (≥99.5 %), sodium bicarbonate ($NaHCO_3$, ≥ 99.5 %) were brought from Shanghai McLean Biochemical Technology Co., Ltd., Shanghai, China. Tetrahydrofuran (THF, anhydrous, ≥ 99.9 %) was brought from Merck & Co., Inc., Darmstadt, Germany. 1,4-dioxane (≥ 99.5 %) was purchased from Tianjin Fuyu Fine Chemical Co., Ltd, Tianjin, China. All purchased chemicals were used as received unless stated otherwise.

2.2. Preparation of methyl p-hydroxycinnamate from lignin

Alkoxide-DMC/THF system: Lignin (80 mg) was added to a Teflon-lined flask containing 6 mmol of an alkali metal alkoxide. The flask was sealed and purged with N_2 gas to maintain an inert atmosphere. Dimethyl carbonate (DMC) and tetrahydrofuran (THF) were then added as the reaction solvents. At predetermined intervals (30 min, 60 min, 90 min, 120 min, 1 d, 3 d, 5 d, 7 d, and 9 d), 1 mL of the reaction mixture was sampled into a glass vial containing 100 μL of a naphthalene internal standard (in DMC) and 50 μL of acetic acid for neutralization. Subsequently, 200 μL of pyridine and 300 μL of Bis(trimethylsilyl)trifluoroacetamide (BSTFA) were added for derivatization at 50 °C for 1 h. The derivatized sample was filtered through a hydrophobic PTFE membrane with a pore size of 0.22 μm and analyzed by gas chromatography-mass spectrometry (GC-MS) and gas chromatography with flame ionization detection (GC-FID).

Alkaline hydrolysis system: Lignin (40 mg) was placed in a 5 mL glass bottle with a magnetic stirrer bar. A 1 mL aliquot of 0.5 mol/L NaOH solution was added, and the sealed vial was stirred in a 30 °C water bath for 6 h. After the reaction, 9 mL of H_2O was added to dilute the reaction mixture. The solution was filtrated through a 0.22 μm hydrophobic PTFE membrane and analyzed directly by liquid chromatography (LC). In a separate procedure, the reaction mixture was neutralized and acidified with 300 μL of 2 mol/L HCl solution. The acidified mixture was concentrated under reduced pressure using a rotary evaporator to remove water. The residue was redissolved in 1 mL THF, followed by the addition of 100 μL naphthalene internal standard (in DMC) Then, 200 μL of pyridine and 300 μL of Bis(trimethylsilyl)trifluoroacetamide (BSTFA) were added derivatization at 50 °C for 1 h in a water bath. After filtrated through a syringe filter, the sample was analyzed by GC-MS.

3. Results and discussion

3.1. Effect of reaction condition on the hydroxycinnamate yield

The effect of different alkali metal alkoxides on the cleavage of ester linkages in herbaceous lignin was investigated under ambient conditions over a period of 3 days. Fig. 1 shows that the use of MeOLi and MeONa primarily led to the formation of *p*CA at 3.4 wt% and 3.3 wt% yield, while MeOK produced hydroxycinnamate, with negligible formation of

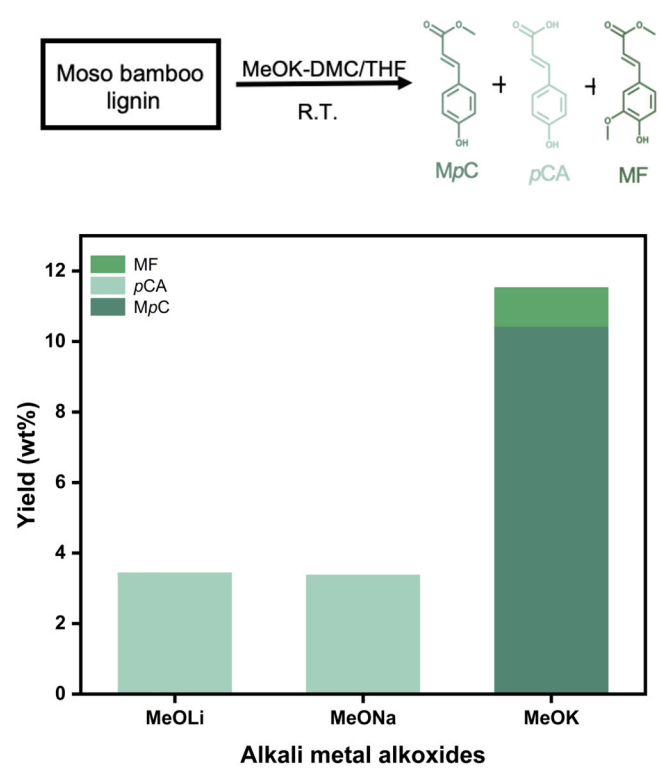

Fig. 1. Reaction of bamboo lignin with different alkali metal alkoxides in aprotic solvent. Reaction conditions: 80 mg of lignin, 6 mmol of alkali metal alkoxide, in 20 mL of DMC/THF (1:3) for 3 d at room temperature.

other products.

Among the tested alkoxides, MeOK demonstrated the highest selectivity toward hydroxycinnamate production. This enhanced performance may be attributed to the greater solubility and dissociation of K^+ ions in the aprotic solvent system. In contrast, the lower dissociation degrees of Na^+ and Li^+ ions in MeONa and MeOLi likely limit their reactivity. According to Christian ionization and dissociation theory, alkali metal alkoxides with larger cationic radii (e.g., $K^+ > Na^+ > Li^+$) exhibit higher mass-to-charge ratio (m/z: K^+=39, Na^+=23, Li^+=7), which correspond to stronger dissociation in aprotic solvents. Greater ionic dissociation enhances the nucleophilicity of the alkoxide, thereby promoting more efficient ester bond cleavage. After 3 days of reaction, the *p*CA yields obtained with MeOLi and MeONa were 3.4 wt% and 3.3 wt%, respectively, both significantly lower than the yield achieved with MeOK. Considering both product yield and selectivity, MeOK is identified as the most effective alkali metal alkoxide for the selective cleavage of ester bonds in lignin.

The impact of different DMC/THF volume ratios on the yield and selectivity of hydroxycinnamic acid and hydroxycinnamate was investigated, as showed in Fig. 2a. When the DMC/THF volume ratio was 1:3, the hydroxycinnamic acid and hydroxycinnamate yield reached 11.5 wt %, indicating that this solvent combination effectively facilitates ester bond cleavage between lignin and MeOK. In contrast, using THF alone (DMC/THF = 0:1) significantly reduced M*p*C yield to 1.2 wt%, while the yields of *p*CA and syringic acid (SA) increased. These results suggest that THF alone is insufficient to promote efficient ester exchange, and the presence of DMC is essential for hydroxycinnamate formation.

When the DMC/THF ratios were adjusted to 1:2 and 1:4, the hydroxycinnamic acid and hydroxycinnamate yield remained comparable to that of 1:3 system. However, a greater number of side products were observed, leading to reduced product selectivity. Therefore, a DMC/THF ratio of 1:3 was identified as the optimal solvent condition for achieving both high yield and product specificity.

Fig. 2b shows the effect of reaction time on M*p*C production. Yields remained low during the initial 120 min but significantly increased after 1 day. Although longer reaction times continued to improve M*p*C yield, the rate of increase slowed, suggesting that sufficient reaction time is necessary to achieve high conversion in this reaction system.

These observations suggest that DMC and THF serve complementary roles in the reaction system. DMC promotes the dissociation of MeOK into MeO^- and K^+, while THF enhances the solubility of lignin and supports nucleophilic activity of MeO^-. A high DMC content favors alkoxide dissociation but limits lignin solubilization. Conversely, an excess of THF improves lignin dispersion but reduces MeO^- availability, limiting nucleophilic attack on ester bonds (Vural et al., 2020, Dabral et al., 2018). Therefore, a DMC/THF volume ratio of 1:3 provides an optimal balance between alkoxide reactivity and lignin solubility, promoting efficient ester cleavage and selective hydroxycinnamic acid and hydroxycinnamate formation.

The influence of reaction temperature on ester bond dissociation of moso bamboo lignin was investigated by comparing reactions conducted at room temperature and at 50 °C. As shown in Fig. 3, the yield of hydroxycinnamate at 50 °C was higher than at room temperature during the initial 120 min. However, after 3 days, hydroxycinnamate yields reached their respective maxima under 50 °C conditions, with no significant improvement upon further extension of reaction time. Interestingly, the formation of *p*CA appeared distinct temperature dependence. At room temperature, *p*CA appeared only after 3 days, whereas at 50 °C, it was already detectable within 120 min. Moreover, a broader array of by-products was observed at the higher temperature.

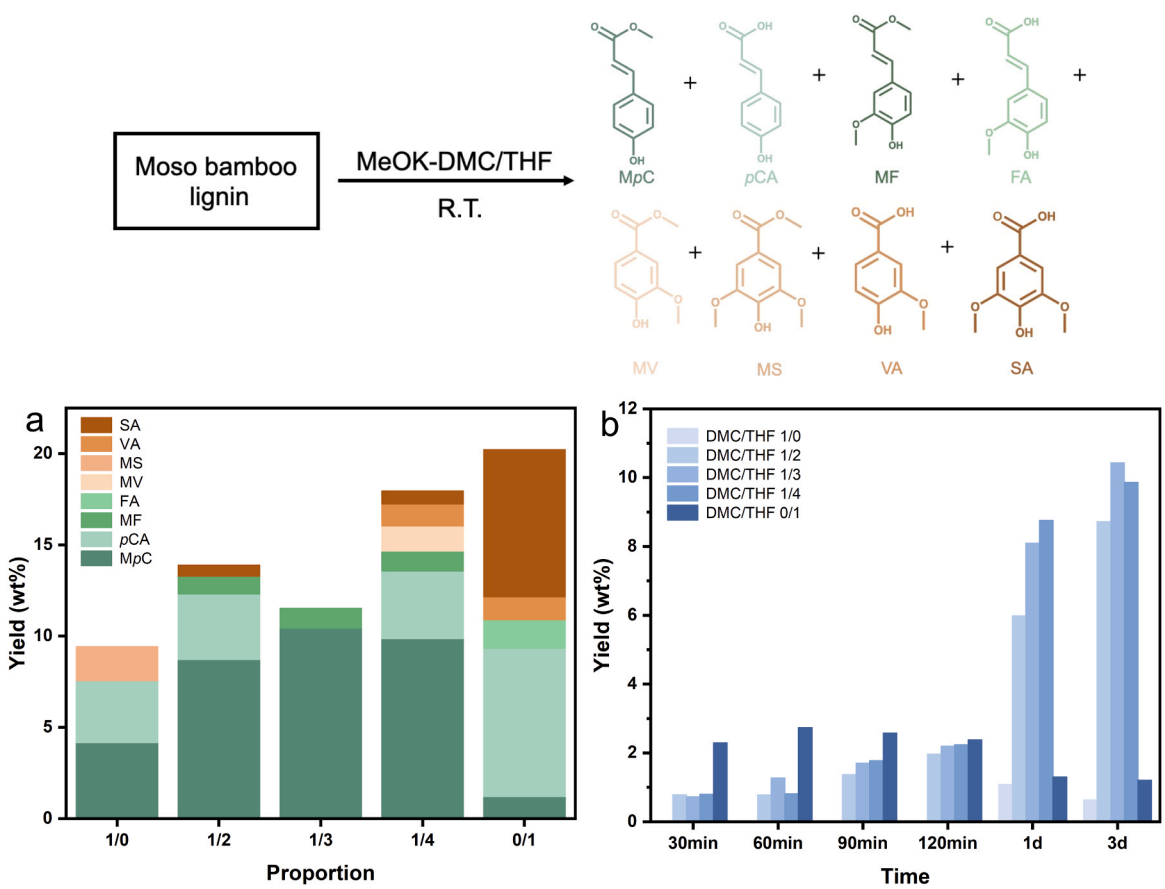

Fig. 2. Effect of reaction solvent ratios (a) and reaction time (b) on the reaction of lignin in DMC/THF. Reaction conditions: 80 mg of lignin, 6 mmol of MeOK, in 20 mL of DMC/THF at room temperature.

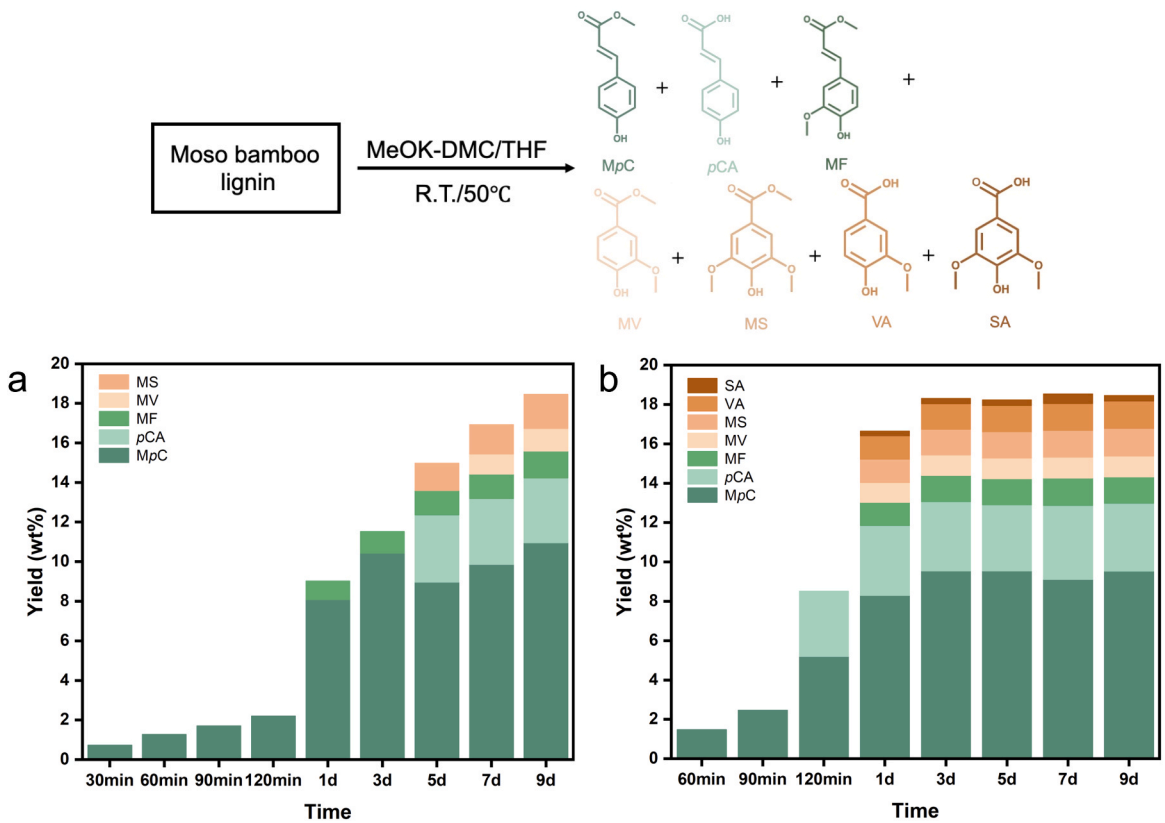

Fig. 3. Reaction of lignin in DMC/THF with MeOK at room temperature (a) and 50 °C (b). Reaction conditions: 80 mg of lignin, 6 mmol of MeOK, in 20 mL of DMC/THF (1:3).

These findings indicate that while temperature has little influence on the final hydroxycinnamic acid and hydroxycinnamate yield, it strongly influences product selectivity. At room temperature, the hydroxycinnamic acid and hydroxycinnamate yield and selectivity were 15.6 wt% and 84.5 %, respectively. At 50 °C, both metrics declined slightly to 14.5 wt% and 78.7 %.

The increased thermal energy at 50 °C enhances molecular motion and intermolecular collisions, thereby activating additional reaction pathways and increasing the complexity of the product mixture. Thus, room temperature conditions are more conducive to achieving high selectivity and a simplified product profile in the depolymerization of lignin to hydroxycinnamic acid and hydroxycinnamate.

3.2. Preparation of hydroxycinnamate from sugarcane bagasse lignin

The results of hydroxycinnamic acid and hydroxycinnamate production from sugarcane bagasse lignin was shown in Fig. 4. The yield of hydroxycinnamic acid and hydroxycinnamate reached 20.8 wt% with a product selectivity of 97.7 %, further confirming the effectiveness of the developed system. The yield peaked after 7 days of reaction, highlighting the critical role of reaction time in maximizing hydroxycinnamic acid and hydroxycinnamate production. The experimental results showed that with the increase of reaction time, the yield of hydroxycinnamic acid and hydroxycinnamate acid in bamboo lignin and bagasse lignin gradually increased. But the rate of increase gradually decreased, reaching its maximum on the seventh and ninth day respectively.

Compared with moso bamboo lignin, sugarcane bagasse lignin produced a higher overall yield of M*p*C under identical reaction conditions. As shown in Fig. 4, during the initial 30–120 min, the yield of *p*CA consistently exceeded that of M*p*C. Over time, the hydroxycinnamic acid yield decreased as hydroxycinnamate became the dominant product.

Syringic acid content gradually decreased throughout the reaction, while MF began to appeared on the first day. Although M*p*C was the major product from both bamboo and bagasse lignin, the two feedstocks exhibited notable differences in the yield and selectivity of by-products. These variations may be attributed to differences in lignin composition and structure between these two species. Overall, sugarcane bagasse lignin demonstrated greater reactivity and conversion efficiency, making it a promising feedstock for the production of hydroxycinnamate. Overall, the MeOK-DMC/THF system was employed to cleave the ester bonds within lignin and yield hydroxycinnamic acid and hydroxycinnamate at room temperature. According to quality balance, 10.0 kg bamboo powder can yield 1.0 kg lignin, can produce 0.16 kg of hydroxycinnamic acid and hydroxycinnamate, and 10.0 kg sugarcane bagasse can yield 1.2 kg lignin and yield 0.25 kg of hydroxycinnamic acid and hydroxycinnamate.

We put 100 mg of lignin into a sealed glass bottle containing 20 mL of 1.0 M NaOH solution and stir at 100 °C for 4 h. The yields of hydroxycinnamic acid groups in bamboo and sugarcane bagasse are 17.2 wt% and 24.3 wt%, respectively. At such condition, most of the esters were cleaved (Zhang et al., 2023) and therefore these yields should be close to theoretical yield of *p*CA and FA in lignin. A comparison between the MeOK-DMC/THF system and conventional alkaline hydrolysis is presented in Table 1. We added bamboo lignin and bagasse lignin to a 1.0 M NaOH solution and reacted at 30 °C for 6 h (Papadimitriou et al., 2024, Li et al., 2019, Zhang et al., 2023, Lu et al., 2023). The yields of hydroxycinnamic acid obtained via alkaline hydrolysis were 6.4 wt% for bamboo lignin and 10.1 wt% for bagasse lignin (Table 2).

Li et al. were prepared a series of metal-based deep eutectic solvents (M-DESs) and used for the catalytic tailoring of lignin H units to produce value-added methyl *p*-hydroxycinnamate (M*p*C) and a yield of 7.8 wt% was obtained (Li et al., 2020). Wu et al. depolymerized bagasse lignin at

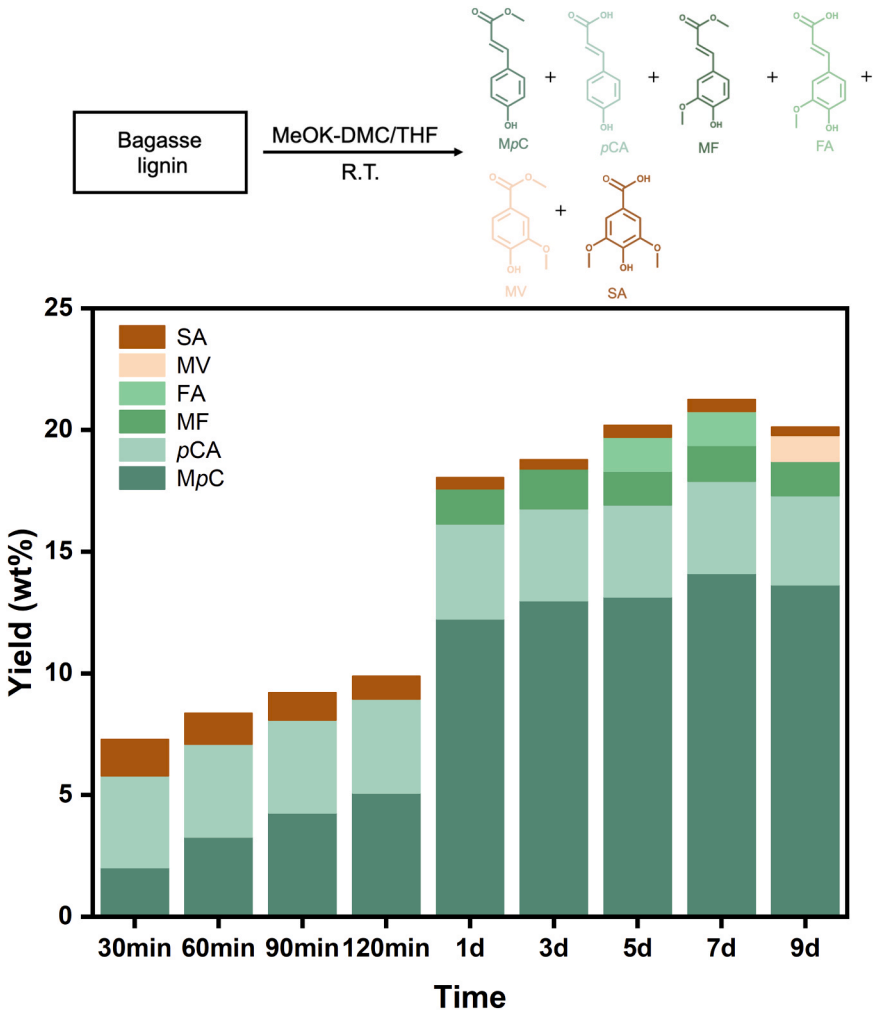

Fig. 4. Hydroxycinnamic acid and hydroxycinnamate yield from bagasse lignin. Reaction conditions: 80 mg of lignin, 6 mmol of MeOK, in 20 mL DMC/THF (1:3) at room temperature.

Table 1
Comparison of the hydroxycinnamic acid and hydroxycinnamate yield using alkaline hydrolysis and MeOK-DMC/THF system.

Lignin raw Material	Method	Yield (wt%) (Hydroxycinnamic acid units)	Yield (wt%) (Hydroxycinnamate units)
Moso bamboo	NaOH 100°C	17.2	-
Moso bamboo	MeOK-DMC/THF	3.2	12.4
Moso bamboo	NaOH 30°C	6.4	-
Sugarcane bagasse	NaOH 100 °C	24.3	-
Sugarcane bagasse	MeOK-DMC/THF	5.2	15.6
Sugarcane bagasse	NaOH 30°C	10.1	-

155 °C for 4 h, the yield of aromatic chemicals was 12.7 %, with a yield of 9.1 % for methyl coumarate (Wu et al., 2022). Goyal et al. isolated 6.55 ± 1.3 mg/unit gram and 9.79 ± 2.1 mg/unit gram of ferulic acid and *p*-coumaric acid from sugarcane bagasse lignin extracted under optimal conditions, respectively (Goyal and Kaur, 2022). By comparison, the MeOK-DMC/THF system achieved a high yield of hydroxycinnamic acid and hydroxycinnamate (15.6 wt% from bamboo lignin and for 20.8 wt% from bagasse lignin) at room temperature. Reached a theoretical yield of 91 % and 85 %, respectively. These results highlight the superior performance of the MeOK-DMC/THF system in terms of both efficiency and selectivity. Its ability to directly convert lignin into value-added chemicals underscores its potential as a promising strategy for sustainable lignin valorization.

Table 2
Abbreviation and full name of the product.

Abbreviation	Full name
*p*CA	*p*-coumarate Acid
M*p*C	Methyl *p*-coumarate
FA	Ferulic Acid
MF	Methyl Ferulic
SA	Syringic Acid
MS	Methyl Syringic
VA	Vanillic Acid
MV	Methyl Vanillic

3.3. HSQC characterization of the lignin

Compared with woody plant lignin, herbaceous lignin contains substantial amount of hydroxycinnamic acid unit, making it a suitable feedstock for the production of hydroxycinnamic acid and

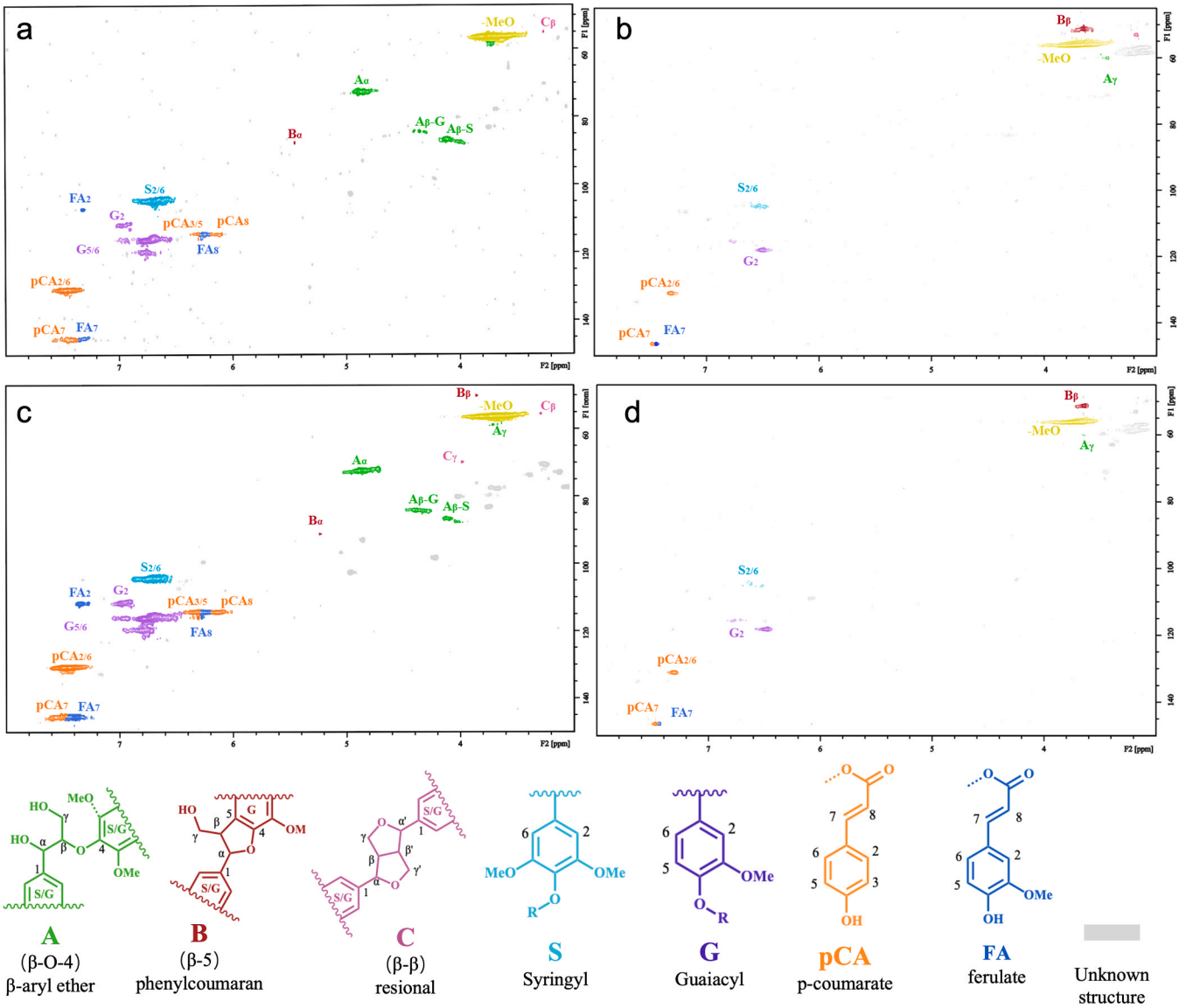

Fig. 5. HSQC spectra of lignin. (a) Bamboo lignin, (b) bamboo lignin after reaction, (c)sugarcane bagasse lignin, and (d) sugarcane bagasse lignin after reaction.

hydroxycinnamate. Fig. 5a and c show the HSQC spectra of lignin extracted from bamboo and sugarcane bagasse using mild acid hydrolysis 1,4-dioxane. In the aromatic region, clear signals corresponding to coniferyl alcohol (G), sinapyl alcohol (S), *p*CA, and FA are observed. (Zhang et al., 2023, Yang See et al., 2021, Wang et al., 2018). In addition, characteristic interunit linkages such as β-O-4, β-β and β-5 are evident in the aliphatic region of the spectra. The changes in the Fig. 5 (b)(d) demonstrate a sharp decrease in the signal of hydroxycinnamic acid structure, with complete disappearance along FA, indicating the cleavage of ester bonds and confirming hydroxycinnamic acid and hydroxycinnamate as the main product. Obviously, the selective cleavage of ester groups in the lignin structure in the MeOK-DMC/THF system is the main reason for obtaining high yields and selectivity of hydroxycinnamic acid and hydroxycinnamate.

3.4. Degradation of model compound in MeOK-DMC/THF

To better understand the reaction mechanism, the reaction of a model compound 3-phenylpropyl cinnamate (3-PPC, 0.06 mmol) was conducted under optimized conditions (a DMC/THF volume ratio of 1:3, total solvent volume of 20 mL, MeOK as the alkali metal alkoxide, room temperature, and a reaction time of 120 min). Under these conditions, the primary cleavage products were methyl 3-phenylpropyl carbonate (M3-PC), originating from the A-ring moiety, and methyl cinnamate (MC), derived from the B-ring moiety. As shown in Fig. 6, partial dissociation of 3-PPC was observed at 10 min. by 20 min, approximately 50 % of the starting material had been cleaved, and at 30 min, the maximum product yields were achieved, with 98.2 % of M3-PC and 85.6 % of MC. These results demonstrate that the system enables rapid and efficient cleavage of 3-PPC. Importantly, both cleavage and methylation occurred, resulting in the formation of methyl esters. This strongly suggests that the reaction cleavage the γ-O ester bond, in which MeO^- attacks the carbonyl of 3-PPC, facilitating product formation. By analogy, during the reaction with lignin, MeO^- similarly targets the ester linkages, particularly those connecting hydroxycinnamic acid units to the lignin, thereby leading to the formation of methylated hydroxycinnamate derivatives such as M*p*C.

4. Conclusion

In this study, a mild and efficient method was successfully developed for the cleavage of ester bond in herbaceous lignin to produce

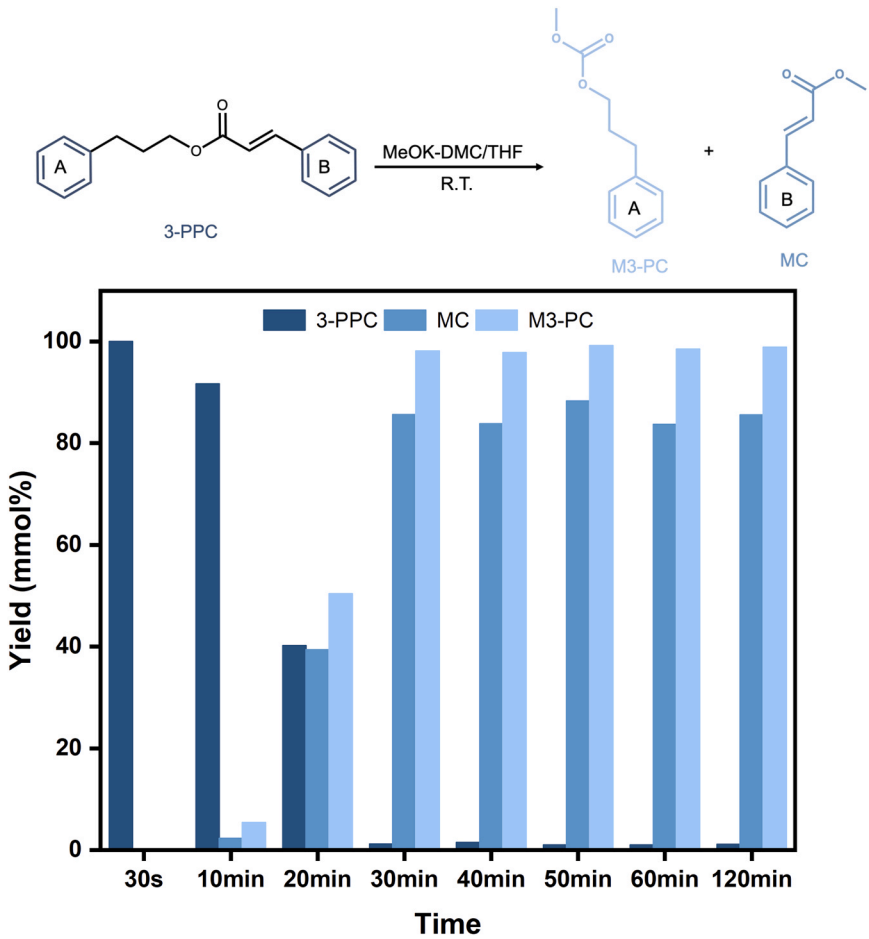

Fig. 6. Reactions of model compound. Reaction conditions: 0.06 mmol of 3-phenylpropyl cinnamate, 3 mmol of MeOK, in 20 mL DMC/THF (1:3) at room temperature.

hydroxycinnamic acid and hydroxycinnamate using a MeOK-DMC/THF system. Key reaction parameters, including solvent composition, type of alkali metal alkoxide, reaction temperature, and lignin feedstock, were systematically investigated to optimize the yield and selectivity of hydroxycinnamic acid and hydroxycinnamate. Under the optimized conditions, moso bamboo lignin yielded 15.6 wt% hydroxycinnamic acid and hydroxycinnamate with a selectivity of 84.5 %, while bagasse lignin achieved a higher yield of 20.8 wt% and selectivity of 97.7 %. The reaction mechanism was elucidated through model compound studies, confirming the cleavage pathways via γ-O ester bond. This approach contributes to the sustainable utilization of biomass and supports the development of green chemical processes within a circular economy framework. However, this system has the problem of excessive time consumption. In the future, we will focus on improving reaction efficiency by using more efficient reactors or optimizing solvent systems.

CRediT authorship contribution statement

Jun Xie: Funding acquisition, Methodology, Resources. **Siqi Huang:** Writing – original draft, Visualization, Validation, Investigation, Data curation. **Xu Zeng:** Validation, Methodology, Funding acquisition, Conceptualization. **Wu Lan:** Writing – review & editing, Resources, Formal analysis. **Qiqi Dai:** Writing – original draft, Visualization, Investigation, Formal analysis. **Minjie Hou:** Visualization, Investigation, Formal analysis. **Chuanfu Liu:** Writing – review & editing, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.122454.

Data availability

Data will be made available on request.

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